

Reinforcement learning algorithms solve Markov decision processes by estimating value functions or optimizing policies from sampled transitions. Two assumptions underlie the standard formulation. First, the agent’s action at time t is determined solely by the observed state s_t and the agent’s policy $\pi(a \mid s_t)$; there is no unobserved variable (confounder) simultaneously influencing both the action and the reward or state transition.¹ Second, the observed state s_t is sufficient for prediction, so that conditioning on s_t renders future states independent of past history. Both assumptions are the Markov property restated in causal language. Both fail routinely in applied settings.

This chapter studies what happens when logged sequential decisions are observational rather than experimental. The central object is the value of a target policy under the interventional law induced by that policy, not merely under the observational law generated by past behavior. The literature now covers confounded MDPs, partial identification and sensitivity analysis, proximal and negative-control methods for POMDPs, instrumental variables, mediator-based identification, and dynamic double machine learning for off-policy evaluation (OPE). The survey articles by [Deng et al. \(2023\)](#) and [da Costa Cunha et al. \(2025\)](#) provide broader coverage of causal representation learning, counterfactual policy optimization, transportability, and fairness.

Table 1 summarizes the main identification strategies. The purpose is not to create a checklist of papers. It is to show economists that many causal RL questions are familiar econometric problems once the estimand is written as a policy value rather than as a static treatment effect.

0.1 From Partial Observability to Causal Structure

A partially observable MDP (POMDP) augments the standard MDP with an observation function. The POMDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, r, \gamma)$, where $\mathcal{O}$ is a finite observation space and O is the observation function.

$$O(o \mid s', a) = P(O_t = o \mid S_t = s', A_{t-1} = a). \quad (1)$$

¹Throughout this chapter, I reserve “outcome” for its causal inference meaning and use the specific RL quantity (reward, state, return, value) elsewhere.

Table 1: Econometric taxonomy for causal reinforcement learning

RL problem	Econometric analogue	Identifying object
Observed state blocks all confounding	Backdoor adjustment, sequential unconfoundedness, g-formula	Interventional transition and reward from conditional models given observed controls
Exogenous variation shifts actions	Instrumental variables and conditional moment restrictions	Structural transition or reward function satisfying moments conditional on instruments
Latent state is not observed but proxies are available	Proximal causal inference and negative controls	Bridge functions that recover policy value from proxy distributions
Actions operate through observed intermediate variables	Front-door, mediation, and g-methods	Mediated transition law and decomposed direct or indirect dynamic effects
No credible point-identifying source exists	Rosenbaum and Tan sensitivity, partial identification	Bounds over policy values under restricted hidden-bias models
High-dimensional nuisance functions are needed	DML, orthogonal scores, efficient OPE	Neyman-orthogonal moments for value or dynamic treatment parameters

The agent does not observe s_t directly but instead receives $o_t \sim O(\cdot \mid s_t, a_{t-1})$ and maintains a belief state $b_t \in \Delta(\mathcal{S})$

$$b_t(s) = P(S_t = s \mid o_1, a_1, \dots, o_t), \quad (2)$$

which is updated via Bayesian filtering at each step. The belief MDP, whose state space is $\Delta(\mathcal{S})$, is itself a fully observable continuous-state MDP, so standard value iteration applies in principle, though computation is intractable in general.

da Costa Cunha et al. (2025) organize sequential decision problems with hidden variables into a hierarchy: standard MDP (full observability, no confounding), POMDP (partial observability, no confounding), confounded MDP (full observability, confounding), and causal POMDP (both). The key distinction is epistemic versus identificational. In a POMDP, the hidden state is a modeling challenge: the agent acknowledges incomplete information and plans accordingly via the belief state, analogous to Kalman or Hamilton filtering in econometrics. In a confounded MDP, the hidden variable is an identification challenge: standard estimators silently produce biased results, analogous to endogeneity and omitted variable bias.

This distinction matters because the same observed sequence can support different econometric interpretations. A hidden patient severity state may be a state variable the physician partially observes, a confounder that drives treatment and prognosis, a source of invalid support for a target policy, or a latent variable for which laboratory histories provide negative controls. Causal RL is the study of which interpretation is credible enough to identify, bound, or estimate the value of a counterfactual policy.

0.2 The Confounded MDP

When unobserved confounders influence both the behavioral policy and the transitions or rewards, the MDP is confounded. This formalization, developed by Zhang and Bareinboim (2019), Zhang and Bareinboim (2020), and Kallus and Zhou (2020), provides the foundation for causal reasoning in sequential decision problems. The key tool is Pearl’s do-operator. The interventional distribution $P(Y \mid \text{do}(X = x))$ is the distribution that arises when X is set externally rather than observed passively, severing all incoming causal influences on X while leaving the remaining data-generating process intact.²

Definition D1 (Confounded MDP (Zhang and Bareinboim, 2020)). *A confounded MDP is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{U}, P, r, \gamma)$ where $\mathcal{S}$ is a finite state space, $\mathcal{A}$ is a finite action space, $\mathcal{U}$ is a space of unobserved confounders, $\gamma \in [0, 1)$ is a discount factor, and the dynamics are governed by structural equations*

$$U_t \sim P_U(\cdot \mid S_t), \quad (3)$$

$$A_t \sim \mu(\cdot \mid S_t, U_t), \quad (4)$$

$$R_t = f_R(S_t, A_t, U_t) + \epsilon_t, \quad (5)$$

$$S_{t+1} \sim f_S(\cdot \mid S_t, A_t, U_t). \quad (6)$$

The behavioral (logging) policy μ depends on the unobserved confounder U_t through Equation (4). An evaluation policy $\pi(a \mid s)$ depends only on the observed state.

Unlike the online algorithms discussed earlier in the survey, where behavior and target

²The do-operator is formalized within the structural causal model (SCM) framework of Pearl (2009). An SCM specifies endogenous variables $\mathbf{V}$, exogenous variables $\mathbf{U}$, structural equations $V_i = f_i(\text{pa}(V_i), U_i)$, and a distribution $P(\mathbf{U})$. The intervention $\text{do}(X = x)$ replaces the structural equation for X with a constant, producing the interventional distribution. See Pearl (2009) for the complete framework, including the causal hierarchy (association, intervention, counterfactual) and general identification theory.

policies were identical or related by a known exploration mechanism, here μ is an unknown function of unobserved variables.

Because μ depends on U_t , conditioning on $\{A_t = a\}$ carries information about the confounder, so the observational and interventional transitions diverge:

$$P(s' | s, a) \neq P(s' | s, \text{do}(a)). \quad (7)$$

The Bellman equation for policy evaluation must use interventional, not observational, transition probabilities. Define the causal Bellman operator for a *target policy* π :

Definition D2 (Causal Bellman Operator (Zhang and Bareinboim, 2020)). *The causal Bellman operator $\mathcal{T}_c^\pi$ for policy π in a confounded MDP is*

$$(\mathcal{T}_c^\pi V)(s) = \sum_{a \in \mathcal{A}} \pi(a | s) \sum_{s' \in \mathcal{S}} P(s' | s, \text{do}(a)) [r(s, a) + \gamma V(s')], \quad (8)$$

where $r(s, a) = \mathbb{E}[R_t | S_t = s, \text{do}(A_t = a)]$ is the interventional expected reward.

Lemma L1 (Bias of Naive Off-Policy Evaluation (Kallus and Zhou, 2020)). *Let $\hat{V}_{naive}^\pi$ denote the value function obtained by solving the Bellman equation with observational transitions $P(s' | s, a)$, and let V^π denote the true value function under interventional transitions $P(s' | s, \text{do}(a))$. In a confounded MDP where $P(s' | s, a) \neq P(s' | s, \text{do}(a))$ for some (s, a, s') , the naive estimator is biased.*

$$\hat{V}_{naive}^\pi(s) \neq V^\pi(s). \quad (9)$$

The importance-sampling estimator is also biased because the propensity $\mu(a | s)$ is not the true behavioral propensity $\mu(a | s, u)$:

$$\hat{V}_{IS}^\pi = \frac{1}{N} \sum_{i=1}^N \prod_{t=0}^T \frac{\pi(a_t^{(i)} | s_t^{(i)})}{\mu(a_t^{(i)} | s_t^{(i)})} G^{(i)}, \quad (10)$$

$$G^{(i)} = \sum_{t=0}^T \gamma^t R_t^{(i)}. \quad (11)$$

Here $G^{(i)}$ is the discounted return of trajectory i and T is the trajectory length.³

³This is the sequential analogue of the omitted variable bias in linear regression. In the static case, regressing Y on X without controlling for a confounder U yields a biased coefficient. In the sequential case, the bias propagates through the Bellman recursion and can amplify over the horizon.

The backdoor criterion (Pearl, 2009) provides a path to identification. Zhang and Bareinboim (2020) apply it to the confounded MDP setting.

Theorem 1 (Backdoor Identification in Confounded MDPs (Pearl, 2009; Zhang and Bareinboim, 2020)). *Suppose a set of observed variables $\mathbf{Z}_t$ satisfies the backdoor criterion relative to (A_t, S_{t+1}) in the causal graph of the confounded MDP, meaning that $\mathbf{Z}_t$ blocks all backdoor paths from A_t to S_{t+1} and no element of $\mathbf{Z}_t$ is a descendant of A_t .⁴ Then the interventional transition probability is identified:*

$$P(s' \mid s, \text{do}(a)) = \sum_{\mathbf{z}} P(s' \mid s, a, \mathbf{z}) P(\mathbf{z} \mid s). \quad (12)$$

Substituting Equation (12) into the causal Bellman operator (Equation 8) yields an identified, consistent estimator of V^π .

0.3 Backdoor-Adjusted Off-Policy Evaluation

Off-policy evaluation under confounding is an average treatment effect estimation problem in a dynamic setting (Bannon et al., 2020): μ is the treatment assignment mechanism, π the counterfactual regime, importance sampling corresponds to inverse probability weighting, and doubly robust OPE corresponds to the AIPW estimator of Robins et al. (1994).

Theorem 1 yields a concrete estimation procedure. Given logged data D_N collected under behavioral policy μ , with elements $(s_t, a_t, z_t, r_t, s_{t+1})$ and observed proxy z_t , the econometrician estimates $\hat{P}(s' \mid s, a, z)$ and $\hat{P}(z \mid s)$ from the logged data, computes

$$\hat{P}(s' \mid s, \text{do}(a)) = \sum_z \hat{P}(s' \mid s, a, z) \hat{P}(z \mid s), \quad (13)$$

and solves the causal Bellman equation using the estimated interventional transitions to obtain $\hat{V}^\pi$. The doubly robust variant combines the fitted action-value function $\hat{Q}(s, a)$ with backdoor-adjusted propensities, achieving consistency if either $\hat{Q}$ or the propensity model is correctly specified.

This is the cleanest case because the analyst observes enough variables to repair

⁴A backdoor path from A_t to S_{t+1} is any path in the causal graph that begins with an arrow into A_t , that is, a non-causal path. In the confounded MDP, $A_t \leftarrow U_t \rightarrow S_{t+1}$ is a backdoor path: U_t causes both A_t and S_{t+1} , creating a spurious association. Blocking all such paths by conditioning on appropriate variables eliminates the confounding bias. See Pearl (2009), Chapter 3.

the state. The rest of the chapter asks what can still be learned when that repair is unavailable, only partially credible, or statistically high-dimensional.

0.4 Alternative Identification Strategies

When no backdoor variable is available, causal RL moves from adjustment to sensitivity analysis, proxies, instruments, mediators, or orthogonal scores. Figure 1 displays the causal graph for three point-identification strategies used in the simulation study.

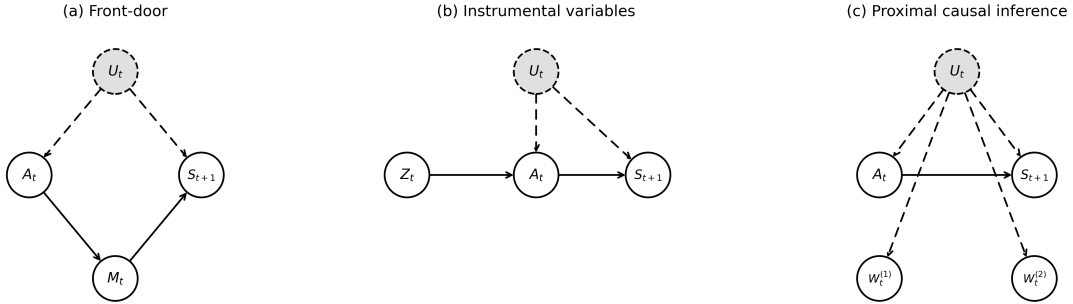


Figure 1: Causal graphs for three identification strategies in confounded MDPs. Gray dashed nodes are unobserved; dashed edges involve unobserved variables. (a) Front-door criterion with mediator M_t . (b) Instrumental variables with exogenous instrument Z_t . (c) Proximal causal inference with proxies $W_t^{(1)}, W_t^{(2)}$.

0.4.1 Sensitivity Analysis and Partial Identification

When point identification is not credible, the estimand becomes a set rather than a number. In static econometrics, this is the logic of Rosenbaum bounds, marginal sensitivity models, and partial identification (Rosenbaum, 2002; Tan, 2006; Manski, 2003). The sequential version asks how much a hidden variable could change the target policy value, allowing the hidden bias to enter each decision epoch.

Kallus and Zhou (2020) study confounding-robust policy evaluation in infinite-horizon RL by bounding the departure between observed and latent propensities. Namkoong et al. (2020) develop related bounds for sequential decisions under unobserved confounding. Bruns-Smith (2021) compare model-free and model-based finite-horizon policy evaluation when causal assumptions are uncertain, showing how worst-case bounds can be propagated through the dynamic problem. Zhang and Bareinboim (2025) extend the Bellman logic further by deriving causal eligibility traces that produce value-function

bounds under hidden confounding and lack of overlap.

This line of work is useful for applied economics because it separates two claims that are often conflated. One claim is statistical: the logged data support a precise estimate under a maintained model. The other is identificational: the maintained model is strong enough to recover the interventional value. Sensitivity methods relax the second claim and report how quickly a policy recommendation changes as hidden bias grows. Recent conformal sensitivity work (Yin et al., 2024) shows the same instinct in static individualized treatment effects, though the RL setting is harder because per-period uncertainty is recursively priced through continuation values.

0.4.2 Proximal and POMDP Bridge Methods

When confounders are latent but proxy variables, noisy correlates of the confounder, are available, proximal causal inference provides identification. Bennett and Kallus (2023) adapt this logic to sequential settings by treating the problem as OPE in a POMDP. The analyst observes two proxies: a treatment-side proxy $W_t^{(1)}$ and an outcome-side proxy $W_t^{(2)}$, both conditionally independent given the latent confounder U_t . Identification proceeds through a bridge function h that solves a conditional moment equation linking the two proxies to the interventional quantity:

$$\mathbb{E}[V(S_{t+1}) \mid W_t^{(1)}, S_t, A_t] = \mathbb{E}[h(W_t^{(2)}, S_t, A_t) \mid W_t^{(1)}, S_t, A_t]. \quad (14)$$

The bridge function h is estimated by solving this integral equation, which reduces to a linear system in the discrete case, and the causal effect is recovered by marginalizing over the outcome proxy distribution.

$$P(s' \mid s, \text{do}(a)) = \sum_{w_2} h(w_2, s, a) P(W^{(2)}=w_2 \mid s). \quad (15)$$

Two bridge functions, analogous to inverse propensity scores and Q-functions, yield a doubly robust estimator of the policy value that is $\sqrt{n}$ -consistent without ever observing U_t . Bennett and Kallus (2023) demonstrate this idea in a sepsis management simulator in which the physician observes a true diabetes status that is only partially recorded in the analyst’s data. Prior clinical observations serve as treatment-side proxies, current clinical observations serve as outcome-side proxies, and the proximal estimator ranks evaluation

policies correctly in between 82% and 100% of test cases where naive estimators fail.

The POMDP bridge literature generalizes this insight beyond tabular examples. Shi et al. (2022) formulate OPE in confounded POMDPs as a minimax learning problem over bridge functions. Uehara et al. (2023) introduce future-dependent value functions that use future proxies as inputs and history proxies as instruments. Hong et al. (2024) move from evaluation to learning by identifying history-dependent policy gradients in confounded POMDPs. Wang et al. (2025) show a related proximal route in which past human or AI actions themselves can carry information about latent decision-relevant states.

0.4.3 Instrumental Variables and Orthogonal Scores

When neither backdoor nor front-door variables are available, instrumental variables can identify causal effects. An instrument Z_t must satisfy relevance, meaning it shifts the action A_t , and an exclusion restriction, meaning its effect on S_{t+1} is channeled entirely through A_t . Liao et al. (2024) formalize this as a Confounded MDP with Instrumental Variables (CMDP-IV), where transitions take the form $S_{t+1} = F^*(S_t, A_t) + \epsilon_t$ with unobserved confounders ϵ_t affecting both the behavior policy and transitions. The transition function F^* is recovered from a conditional moment restriction:

$$\mathbb{E}[S_{t+1} - F^*(S_t, A_t) \mid Z_t, S_t] = 0. \quad (16)$$

For a binary action and binary instrument, the Wald estimator provides a closed-form solution. Let β denote the causal effect of promoting (action $a = 0$) on the transition probability.

$$\beta = \frac{P(s' \mid s, Z=1) - P(s' \mid s, Z=0)}{P(A=0 \mid s, Z=1) - P(A=0 \mid s, Z=0)}, \quad (17)$$

where the numerator is the reduced-form effect of the instrument on the transition and the denominator is the first-stage effect on treatment uptake. The interventional transition is then recovered from any instrument value z via $P(s' \mid s, \text{do}(a=0)) = P(s' \mid s, Z=z) + \beta \cdot (1 - P(A=0 \mid s, Z=z))$. Their IV-aided Value Iteration algorithm applies this moment restriction at each state to estimate F^* , then runs standard value iteration on the estimated model.

Liao et al. (2024) illustrate with neonatal intensive care unit (NICU) assignment. A hospital must decide whether to admit each newborn to the NICU, and this decision is confounded by unobserved severity indicators that affect both admission and infant health. Differential travel time to a specialty care provider serves as an instrument. Relevance holds because travel time shifts referral probabilities. The exclusion restriction requires that travel time affect infant health only through admission, not through any other channel.

Orthogonal-score methods solve a different problem. They do not by themselves remove hidden confounding, but they make causal OPE statistically stable when the identifying assumptions reduce the value to nuisance functions that must be learned flexibly. Liao et al. (2021) estimate long-term average policy values for mobile health applications. Lewis and Syrgkanis (2021) extend double machine learning to dynamic treatment effects through sequential residualization and Neyman-orthogonal moments. Kallus and Uehara (2022) derive efficient double reinforcement learning estimators for Markovian OPE using q-functions and stationary density ratios. The common econometric message is that identification and estimation are separate: first specify why the policy value is identified, then use orthogonal scores to keep first-stage machine learning error from dominating the target estimate.

0.4.4 Mediator-Based Identification

The front-door criterion applies when A_t affects S_{t+1} only through an observed mediator M_t . Three conditions are required: M_t intercepts all directed paths from A_t to S_{t+1} , no unblocked backdoor path exists from A_t to M_t , and A_t blocks all backdoor paths from M_t to S_{t+1} . When satisfied (Pearl, 2009),

$$P(s' | s, \text{do}(a)) = \sum_m P(m | a) \sum_{a'} P(s' | s, m, a') P(a' | s). \quad (18)$$

The first factor is unconfounded; the inner sum adjusts for confounding on the $M_t \rightarrow S_{t+1}$ link by averaging over the observational action distribution. This is the sequential analogue of mediation analysis.

Mediator-based RL is most attractive when the action changes an observable channel before it changes the long-run state. In mobile health, an app prompt may affect corti-

solved through supplement adherence. In retail pricing, a promotion may affect conversion through a marketing follow-up or browsing response. [Shi et al. \(2024\)](#) use mediator variables to identify and construct confidence intervals for policy values in confounded MDPs. [Ge et al. \(2023\)](#) develop an RL framework for dynamic mediation analysis, decomposing the long-run effect into direct and mediated components. The policy question shifts from “does the action work” to “through which sequential channel does the action work, and is that channel stable enough to transport across policies?”

0.5 The Broader Causal RL Landscape

The first broader direction is nonidentifiability under hidden confounding. In many offline RL problems, no observed control, instrument, mediator, or proxy is strong enough to point-identify the value of a new policy. [Pace et al. \(2024\)](#) respond to this problem with Delphic offline RL, which treats hidden confounding as a source of nonidentifiable uncertainty rather than pretending it has been solved. This perspective is close to robust decision making in economics: the agent optimizes while carrying a set of plausible worlds, not a single estimated MDP.

The second direction is counterfactual credit assignment and causal imitation learning. [Buesing et al. \(2019\)](#) introduce Gumbel-Max structural causal models that produce counterfactual rollouts from a single observed trajectory via Pearl’s Abduction-Action-Prediction procedure ([Pearl, 2009](#)): infer the exogenous noise explaining the observed trajectory, replace the logged action, and propagate through the structural equations. [Oberst and Sontag \(2019\)](#) extend this to healthcare settings, while [Mesnard et al. \(2021\)](#) develop model-free counterfactual credit assignment by conditioning value functions on future information that is constrained not to reveal the agent’s own action. Causal imitation learning uses the same instinct in demonstration data: a learner should imitate the causal decision rule, not merely the observed association between histories and expert actions.

The third direction is causal transportability and invariance. Transportability theory ([Pearl and Bareinboim, 2014](#); [Bareinboim and Pearl, 2016](#)) asks which mechanisms are stable across source and target environments. In RL, this governs sim-to-real transfer, curriculum learning, and policy adaptation under environment shift. [Li et al. \(2024\)](#) show that curriculum tasks can be causally misaligned when unobserved confounders

break the invariance of optimal decision rules, and they derive graphical conditions for aligned source tasks. This brings causal representation learning and transfer back to an econometric question: which structural mechanisms are invariant enough to justify pooling data across environments?

0.6 Simulation Study: Confounded Retail Pricing MDP

I construct a 5-state engagement funnel $\mathcal{S} = \{0, 1, 2, 3, 4\}$ with two actions (promote, hold price) and an absorbing conversion state at $s = 4$. A retailer manages customers through engagement stages, deciding whether to offer a promotional discount. The data-generating process embeds four distinct sources of causal variation, enabling simultaneous validation of four point-identification strategies from a single DGP.

Figure 2 displays the complete causal graph. Market conditions $Z_t \sim \text{Bernoulli}(0.5)$ are observed and affect both the latent confounder and transitions. Consumer sentiment U_t is an unobserved confounder strongly correlated with Z_t .⁵ An independent cost shock $IV_t \sim \text{Bernoulli}(0.5)$ serves as an instrument. The behavioral pricing policy depends on both U_t and IV_t : $\mu(\text{promote} \mid s, U_t, IV_t) = 0.55 + \rho \cdot 0.25 \cdot (2U_t - 1) + 0.15 \cdot (IV_t - 0.5)$, where $\rho \in \{0, 0.2, \dots, 1.0\}$ controls confounding strength. Promotions trigger marketing follow-ups with M_t serving as a mediator.⁶ Two noisy proxies of U_t are available: a CRM score $W_t^{(1)}$ and browsing behavior $W_t^{(2)}$.⁷

The action affects the next state only through the mediator: $P(s+1 \mid s, M_t, Z_t)$ depends on M_t and Z_t but not on A_t or U_t directly. This enables all four identification strategies simultaneously: Z_t satisfies the backdoor criterion, M_t satisfies the front-door criterion, IV_t satisfies relevance and exclusion, and $W_t^{(1)}, W_t^{(2)}$ satisfy the proximal conditions.⁸

I compare six estimators: oracle, naive (biased per Lemma L1), backdoor (Equation 13), front-door (Equation 18), Wald IV (Equation 17), and proximal (Equations 14 and 15).⁹ These estimators correspond to the point-identification rows of Table 1. The simulation intentionally leaves out sensitivity bounds and orthogonal-score refinements

⁵ $P(U_t=1 \mid Z_t=1) = 0.9$ and $P(U_t=1 \mid Z_t=0) = 0.1$.

⁶ $M_t \sim \text{Bernoulli}(0.8)$ when the retailer promotes and $\text{Bernoulli}(0.2)$ otherwise.

⁷ $W_t^{(1)} \sim \text{Bernoulli}(0.85 \cdot U_t + 0.15 \cdot (1 - U_t))$ and $W_t^{(2)} \sim \text{Bernoulli}(0.75 \cdot U_t + 0.25 \cdot (1 - U_t))$.

⁸Rewards are $r(s, a) = -1$ for $s < 4$, $\gamma = 0.9$. The target policy always promotes. The true interventional transition probability is $P(s+1 \mid s, \text{do}(\text{promote})) = 0.615$.

⁹Each configuration uses 2,000 trajectories averaged over 20 seeds.

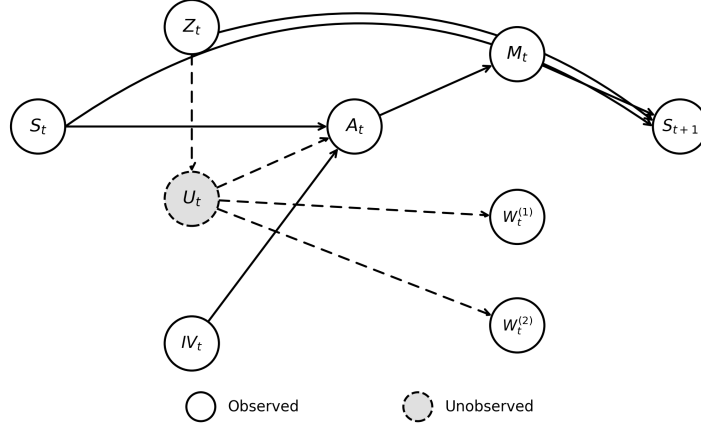


Figure 2: Complete causal graph of the simulation DGP. Gray dashed node (U_t) is unobserved; dashed edges involve U_t .

so that the figure isolates the identification logic from higher-order statistical efficiency.

Figure 3 reports bias and RMSE across confounding strengths (20 seeds, 2,000 trajectories per seed). The naive estimator’s bias grows monotonically with ρ because observational transitions overestimate promotion success: the promote action is more likely when $U_t = 1$, which correlates with favorable conditions, so $\hat{P}_{\text{obs}}$ exceeds the true interventional probability, and this per-step bias compounds through the Bellman recursion. The backdoor and front-door estimators eliminate bias at all ρ , validating Theorem 1 and Equation (18). The IV estimator maintains low bias but higher variance due to the Wald ratio’s sensitivity to instrument strength; panel (c) illustrates this classic bias-variance tradeoff. The proximal estimator achieves low bias with moderate variance, confirming that bridge functions recover causal effects from noisy proxies.

A final bridge to structural econometrics is worth making explicit. [Adusumilli and Eckardt \(2025\)](#) use temporal-difference learning to estimate continuation-value terms in dynamic discrete choice models, linking RL-style Bellman residuals to the conditional choice probability tradition after [Rust \(1987\)](#) and [Hotz and Miller \(1993\)](#). That work is not primarily about hidden confounding, but it shows why causal RL and structural estimation are converging: both fields estimate counterfactual values in dynamic systems, and both must decide which parts of the observed transition law are policy-invariant

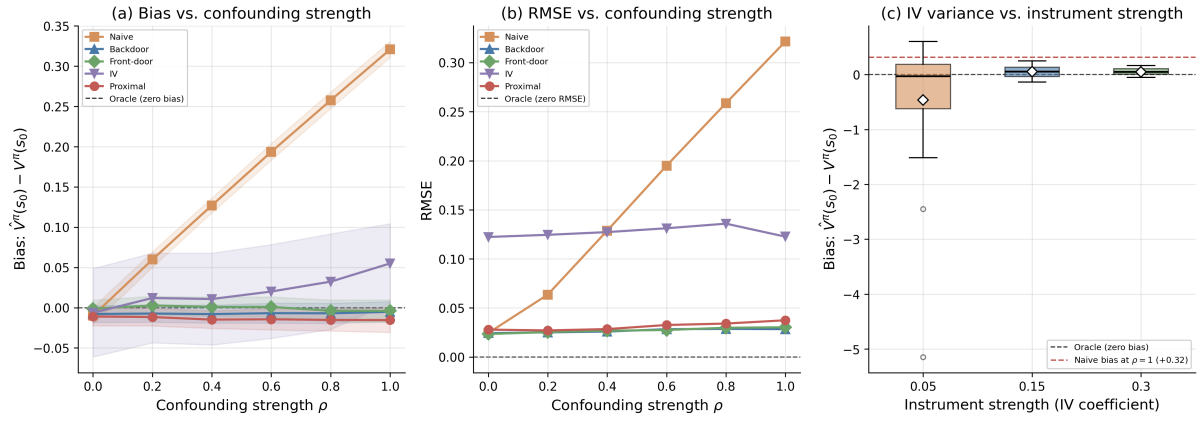


Figure 3: (a) Bias of five OPE estimators as a function of confounding strength ρ . (b) RMSE of five estimators as a function of ρ . (c) IV estimator bias distribution vs. instrument strength at $\rho = 1$; dashed red line is naive estimator bias.

structural objects.

References

- Karun Adusumilli and Dita Eckardt. Temporal-difference estimation of dynamic discrete choice models. *The Review of Economic Studies*, page rdaf081, 2025. doi: 10.1093/restud/rdaf081.
- James Bannon, Brad Langlois, Raimundo Fernandez, and Danielle Maddix. Causality and batch reinforcement learning: Complementary approaches to planning in unknown domains. In *NeurIPS Workshop on Causal Discovery and Causality-Inspired Machine Learning*, 2020.
- Elias Bareinboim and Judea Pearl. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352, 2016.
- Andrew Bennett and Nathan Kallus. Proximal reinforcement learning: Efficient off-policy evaluation in partially observed Markov decision processes. *Operations Research*, 72(3):1071–1086, 2023. doi: 10.1287/opre.2021.0781.
- David A. Bruns-Smith. Model-free and model-based policy evaluation when causality is uncertain. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 1116–1126. PMLR, 2021.
- Lars Buesing, Theophane Weber, Yori Zwols, Sebastien Racaniere, Arthur Guez, Jean-Baptiste Lespiau, and Nicolas Heess. Woulda, coulda, shoulda: Counterfactually-guided policy search. In *International Conference on Learning Representations*, 2019.
- Cristiano da Costa Cunha, Wei Liu, Tim French, and Ajmal Mian. Unifying causal reinforcement learning: Survey, taxonomy, algorithms and applications. *arXiv preprint arXiv:2512.18135*, 2025.
- Zhihong Deng, Jing Jiang, Guodong Long, and Chengqi Zhang. Causal reinforcement learning: A survey. *Transactions on Machine Learning Research*, 2023.
- Lin Ge, Jitao Wang, Chengchun Shi, Zhenke Wu, and Rui Song. A reinforcement learning framework for dynamic mediation analysis. In *Proceedings of the 40th International*

- Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 11050–11097. PMLR, 2023.
- Mao Hong, Zhengling Qi, and Yanxun Xu. A policy gradient method for confounded POMDPs. In *International Conference on Learning Representations*, 2024.
- V. Joseph Hotz and Robert A. Miller. Conditional choice probabilities and the estimation of dynamic models. *The Review of Economic Studies*, 60(3):497–529, 1993.
- Nathan Kallus and Masatoshi Uehara. Efficiently breaking the curse of horizon in off-policy evaluation with double reinforcement learning. *Operations Research*, 70(6):3282–3302, 2022. doi: 10.1287/opre.2021.2249.
- Nathan Kallus and Angela Zhou. Confounding-robust policy evaluation in infinite-horizon reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- Greg Lewis and Vasilis Syrgkanis. Double/debiased machine learning for dynamic treatment effects. In *Advances in Neural Information Processing Systems*, volume 34, pages 22695–22707, 2021.
- Mingxuan Li, Junzhe Zhang, and Elias Bareinboim. Causally aligned curriculum learning. In *International Conference on Learning Representations*, 2024.
- Luofeng Liao, Zuyue Fu, Zhuoran Yang, Yixin Wang, Dingli Ma, Mladen Kolar, and Zhaoran Wang. Instrumental variable value iteration for causal offline reinforcement learning. *Journal of Machine Learning Research*, 25(303):1–56, 2024.
- Peng Liao, Predrag Klasnja, and Susan A. Murphy. Off-policy estimation of long-term average outcomes with applications to mobile health. *Journal of the American Statistical Association*, 116(533):382–391, 2021. doi: 10.1080/01621459.2020.1807993.
- Charles F. Manski. *Partial Identification of Probability Distributions*. Springer, New York, 2003.
- Thomas Mesnard, Theophane Weber, Fabio Viola, Shantanu Thakoor, Alaa Saade, Anna Harutyunyan, Will Dabney, Thomas S. Stepleton, Nicolas Heess, Arthur Guez, Eric

- Moulines, Marcus Hutter, Lars Buesing, and Remi Munos. Counterfactual credit assignment in model-free reinforcement learning. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 7654–7664. PMLR, 2021.
- Hongseok Namkoong, Ramtin Keramati, Steve Yadlowsky, and Emma Brunskill. Off-policy policy evaluation for sequential decisions under unobserved confounding. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- Michael Oberst and David Sontag. Counterfactual off-policy evaluation with Gumbel-Max structural causal models. In *Proceedings of the 36th International Conference on Machine Learning*, pages 4923–4932, 2019.
- Alizée Pace, Hugo Yèche, David Ha, Gunnar Rätsch, and Guy Tennenholtz. Delphic offline reinforcement learning under nonidentifiable hidden confounding. In *International Conference on Learning Representations*, 2024.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition, 2009.
- Judea Pearl and Elias Bareinboim. External validity: From do-calculus to transportability across populations. *Statistical Science*, 29(4):579–595, 2014.
- James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994.
- Paul R. Rosenbaum. *Observational Studies*. Springer, New York, 2 edition, 2002. doi: 10.1007/978-1-4757-3692-2.
- John Rust. Optimal replacement of gmc bus engines: An empirical model of harold zurcher. *Econometrica*, 55(5):999–1033, 1987.
- Chengchun Shi, Masatoshi Uehara, Jiawei Huang, and Nan Jiang. A minimax learning approach to off-policy evaluation in confounded partially observable markov decision processes. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 20057–20094. PMLR, 2022.

- Chengchun Shi, Jin Zhu, Ye Shen, Shikai Luo, Hongtu Zhu, and Rui Song. Off-policy confidence interval estimation with confounded markov decision process. *Journal of the American Statistical Association*, 119(545):273–284, 2024. doi: 10.1080/01621459.2022.2110878.
- Zhiqiang Tan. A distributional approach for causal inference using propensity scores. *Journal of the American Statistical Association*, 101(476):1619–1637, 2006. doi: 10.1198/016214506000000023.
- Masatoshi Uehara, Haruka Kiyohara, Andrew Bennett, Victor Chernozhukov, Nan Jiang, Nathan Kallus, Chengchun Shi, and Wen Sun. Future-dependent value-based off-policy evaluation in POMDPs. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Jiayi Wang, Zhengling Qi, and Chengchun Shi. Blessing from human-AI interaction: Super policy learning in confounded environments. *Journal of the American Statistical Association*, 2025. doi: 10.1080/01621459.2025.2574706. In press.
- Mingzhang Yin, Claudia Shi, Yixin Wang, and David M. Blei. Conformal sensitivity analysis for individual treatment effects. *Journal of the American Statistical Association*, 119(545):122–135, 2024. doi: 10.1080/01621459.2022.2102503.
- Junzhe Zhang and Elias Bareinboim. Near-optimal reinforcement learning in dynamic treatment regimes. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Junzhe Zhang and Elias Bareinboim. Designing optimal dynamic treatment regimes: A causal reinforcement learning approach. In *Proceedings of the 37th International Conference on Machine Learning*, pages 11012–11022, 2020.
- Junzhe Zhang and Elias Bareinboim. Causal eligibility traces for confounding robust off-policy evaluation. In *Proceedings of the Forty-First Conference on Uncertainty in Artificial Intelligence*, volume 286 of *Proceedings of Machine Learning Research*, pages 4933–4942. PMLR, 2025.